

Module Quiz and Answer

1. Which Python method is used to convert all characters in a string to lowercase?

- a) string.upper()
- b) string.casefold()
- c) string.title()
- d) string.strip()

Ans: b) string.casefold()

ব্যখ্যা: casefold() এবং lower() দুটোই lowercase করে, কিন্তু casefold() Unicode characters-এর জন্য বেশি reliable। তবে সবচেয়ে common answer হলো casefold()। Learning: casefold() is preferred for case-insensitive text normalization।

2. Which Python library is commonly used to remove HTML tags from text?

- a) re
- b) BeautifulSoup
- c) nltk
- d) spacy

Ans: b) BeautifulSoup

ব্যখ্যা: BeautifulSoup একটা HTML parsing library যেটা দিয়ে HTML tags সহজে remove করা যায়। re দিয়েও সম্ভব কিন্তু সেটা reliable না complex HTML-এর জন্য। Learning: BeautifulSoup's get_text() method extracts clean text from HTML।

3. What does the following regex pattern match? r"https?://S+"

- a) Only HTTP URLs
- b) Only HTTPS URLs
- c) Both HTTP and HTTPS URLs
- d) All special characters in a URL

Ans: c) Both HTTP and HTTPS URLs

ব্যখ্যা: https? মানে 's' টা optional — তাই এটা http এবং https দুটোই match করে। \S+ মানে whitespace ছাড়া যেকোনো character। Learning: The ? quantifier makes the preceding character optional in regex।

4. Which of the following is NOT considered a stopword in English?

- a) the
- b) is
- c) machine
- d) and

Ans: c) machine

ব্যখ্যা: Stopwords হলো সেই common words যেগুলো sentence-এ বেশি থাকে কিন্তু meaning carry করে না — যেমন the, is, and। "machine" একটা meaningful content word। Learning: Stopwords are removed to reduce noise and focus on meaningful tokens।

5. What is the main difference between Stemming and Lemmatization?

- a) Stemming uses a dictionary, Lemmatization does not
- b) Lemmatization uses a dictionary and grammar rules, Stemming uses rule-based cutting
- c) Both produce the same output
- d) Stemming is slower than Lemmatization

Ans: b) Lemmatization uses a dictionary and grammar rules, Stemming uses rule-based cutting

ব্যখ্যা: Stemming শুধু suffix কেটে দেয় — ফলে কখনো কখনো valid word পাওয়া যায় না। Lemmatization dictionary এবং grammar বুঝে সঠিক base form দেয়। Learning: Lemmatization is more accurate but slower; stemming is faster but less precise।

6. What is the output of the following code?

```
import nltk
nltk.word_tokenize("I can't stop.")
```

- a) ["I", "can't", "stop", "."]
- b) ["I", "can", "t", "stop"]
- c) ["I", "ca", "n't", "stop", "."]
- d) ["I", "cannot", "stop", "."]

Ans: c) ["I", "ca", "n't", "stop", "."]

ব্যখ্যা: nltk.word_tokenize() contractions কে নিজের নিয়মে split করে। "can't" কে এটা "ca" এবং "n't" হিসেবে tokenize করে — এটা NLTK-এর known behavior। Learning: nltk.word_tokenize() splits contractions based on Punkt tokenizer rules, not simple apostrophe splitting।

7. What is the output of PorterStemmer().stem("running")?

- a) running
- b) runner
- c) run
- d) runn

Ans: c) run

ব্যখ্যা: PorterStemmer একটা rule-based stemmer যেটা "running" থেকে "ing" suffix কেটে "run" দেয়। এটা linguistically সঠিক কারণ "run" হলো root form। Learning: Porter Stemmer applies a series of rules to progressively strip suffixes।

8. Which of the following is the correct way to remove punctuation from text using Python's string library?

- a) `text.replace(string.punctuation, "")`
- b) `"".join([ch for ch in text if ch not in string.punctuation])`
- c) `text.strip(string.punctuation)`
- d) `string.punctuation.remove(text)`

Ans: b) `"".join([ch for ch in text if ch not in string.punctuation])`

ব্যখ্যা: `string.punctuation` এ সব punctuation characters আছে। List comprehension দিয়ে প্রতিটা character check করে punctuation বাদ দিয়ে join করা হয়। এটা সবচেয়ে common এবং reliable approach। Learning: Iterating over characters and filtering using `string.punctuation` is the standard way to remove punctuation।

9. Which tokenizer correctly handles the contraction "don't" by splitting it into ["do", "n't"]?

- a) `str.split()`
- b) `nltk.word_tokenize()`
- c) `text.split(" ")`
- d) `re.split(r'\s+', text)`

Ans: b) `nltk.word_tokenize()`

ব্যখ্যা: Simple `split()` শুধু whitespace দিয়ে ভাগ করে — "don't" কে একটা token হিসেবে রাখে। `nltk.word_tokenize()` language rules বোঝে এবং contractions সঠিকভাবে split করতে পারে। Learning: `nltk.word_tokenize()` uses a pre-trained Punkt tokenizer that handles contractions and punctuation intelligently।

10. What does lemmatization return for the word "better" when `pos="a"` (adjective) is specified in `WordNetLemmatizer`?

- a) better
- b) best
- c) good
- d) well

Ans: c) good

ব্যখ্যা: `WordNetLemmatizer` জানে যে "better" হলো "good" এর comparative form। `pos="a"` (adjective) specify করলে এটা সঠিকভাবে "good" return করে। `pos` না দিলে "better" ই থাকে। Learning: Specifying the correct POS tag in lemmatization produces accurate base forms।